

How many starts below x halt? The Space Needle, counted

WS2, first execution — an unconditional depth-graded density theorem — July 26, 2026

1. Result

Let $A_j(x)$ count the $b \leq x$ whose orbit first reaches a power of 2 after exactly j steps. The workstream asked for $\#\{b \leq x : b \text{ halts}\} = O(x^c)$ with $c < 1$. What is actually provable, and is proved here, is sharper in one direction and weaker in another:

Theorem. For every L , $\sum_{j \leq L} A_j(x) \leq (L+1) \cdot (\log_2 x + 1.585L + 2)^{L+1}$.

Sharper: at bounded depth the halting starts are **polylogarithmic**, not merely of density x^c . Weaker: it says nothing at unbounded depth, and Section 5 explains why that gap is the cryptid difficulty itself rather than a missing lemma. The proof needs only two facts about the map, both proved below, and it comes with an exact computation: the depth-graded backward enumeration is **complete** (no ceiling assumption), so the true counts can be evaluated at astronomical x — up to 10^{192} in a fraction of a second — and they grow like $(\log_2 x)^{1.0}$, about 1.7 seeds per octave.

2. A premise of the plan, corrected

NEXT_STEPS.md justified this workstream with “the Needle map is strictly increasing, so H is exactly backward-enumerable”. The map is **not** increasing: $F(9) = 21 > F(10) = 17$, and there are 990 inversions below 2000. The property the argument actually needs is **expansion** — each step strictly grows its argument — which is true and is what makes preimages smaller than their images, hence the backward walk finite. Everything downstream survives the correction; the plan text has been fixed.

3. The three lemmas

lemma	statement and proof
L1. Expansion	$x + 3 \leq F(x) \leq 2.5x + v \leq 3x$ for every $x \geq 3$ that is not a power of 2. <i>Proof:</i> $F(x) = x + 3k + v$ with $k = x \gg (v+1) \geq 1$, giving the lower bound; and $k \leq x/2$ with $v \leq \text{floor}(\log_2 x) \leq x/2$ gives the upper. Machine-verified for $x < 200,000$.
L2. Exact backward step	$y = F(b)$ with $v_2(b) = v$ holds iff $b = (2^{v+1}y + 2^v(3 - 2v)) / (2^{v+1} + 3)$. So each valuation contributes <i>at most one</i> preimage, only $v \leq \log_2(2y)$ can contribute, and therefore $d(y) := \#F^{-1}(y) \leq \log_2 y + 1$. Machine-verified: the formula inverts F for all $x < 60,000$, and the largest $d(y)$ seen below 60,000 is 4.
L3. Subcritical branching on average	$\sum_{y \leq Y} d(y) \leq cY + O(\log Y)$ with $c = \sum_{v \geq 0} 1/(2^{v+1} + 3) = 0.5453... < 1$, because branch v produces a preimage only for y in one residue class mod $2^{v+1} + 3$. Measured average over $y < 200,000$: 0.5452 — the rigorous ceiling is essentially attained, so the backward tree is thin for a reason that can be written down, not just observed.

lemma	statement and proof
L4. Completeness (the cutoff lemma)	If $b \leq x$ and $F^j(b) = 2^m$ with $j \leq L$, then $2^m \leq 3^L x$ by L1. So seeding the backward walk with every power of 2 up to $3^L x$, and pruning layer i at $3^{L-i} x$, enumerates every such b and no others. This is what replaces the heuristic ceiling ($CEIL = 2^{44}$) in <code>explorations/backward.py</code> : the counts below are theorems about the map, not about a search budget.

The theorem follows: $A_0(x) \leq \log_2 x + 1$, and a depth- $(j+1)$ seed below x is a preimage of a depth- j seed below $3x$ (L1), of which there are at most $\log_2(3x) + 1$ per node (L2), so $A_{j+1}(x) \leq A_j(3x) \cdot (\log_2(3x) + 1)$; unwinding gives the stated bound.

4. The exact counts

x	$A_0 \dots A_5$	total (depth ≤ 5)	the theorem's bound	total / $\log_2 x$
10^6	20 9 5 1 0 0	35	4.3e+09	1.76
10^{12}	40 18 8 2 2 2	72	9.1e+10	1.81
10^{24}	80 34 14 5 3 5	141	3.1e+12	1.77
10^{48}	160 71 26 8 5 7	277	1.4e+14	1.74
10^{96}	319 141 52 20 13 13	558	7.6e+15	1.75
10^{192}	638 276 105 43 25 21	1108	4.4e+17	1.74

Every entry is exact and complete by L4, and the whole table takes 0.3 seconds. Consecutive rows give a growth exponent in $\log_2 x$ of 1.04, 0.97, 0.97, 1.01, 0.99 — so the depth-bounded halting seeds grow **linearly in $\log x$** , about 1.7 per octave, which is the exact form of the “one halting seed per octave” observation (Finding 5) that was previously only empirical. The rigorous bound is loose by many orders of magnitude, as expected: it pays $\log_2 x$ per level of depth where the truth pays a factor of about 0.43.

Two structural facts fall out of the same computation. The layer sizes decay geometrically in depth (at $x = 10^{192}$: 638, 276, 105, 43, 25, 21), which is L3's subcriticality visible in the tree rather than on average. And the tree is genuinely shallow at small x : below 10^6 the depth-14 enumeration finds exactly the same 35 seeds as the depth-3 one — 20 powers of 2 and 15 non-trivial seeds, the smallest being 7, 103, 312, 352, 372.

5. What is not proved, and why that is the interesting part

The bound degrades as depth grows and goes trivial at depth about $\log x / \log \log x$. To close the gap to an unconditional $O(x^c)$ for *all* depths one would need the branching to stay subcritical along the tree, not merely on average over all integers — i.e. that the tree's nodes are equidistributed among the residue classes mod $2^{v+1}+3$ that L3 counts. That is precisely the equidistribution-of-one-orbit wall that makes these machines cryptids: the same gap as Collatz's, in the same place. It is worth being explicit that this is not a missing technical lemma. An unconditional $O(x^c)$ with $c < 1$ would assert that all but x^c starts never halt, a statement about individual orbits of every start — strictly stronger than any almost-everywhere result the field currently has for any expanding Collatz-like map.

The productive reading is that depth is the right grading, and it is the same grading WS3 uses for the Baker ladder (block count). The two workstreams meet: WS2 bounds how many seeds can halt within L steps, WS3 aims to exclude halting for branch words of bounded complexity. A combined statement — nothing of bounded depth halts, and only polylog many things of bounded depth exist — is the realistic shape of an unconditional partial decision for this machine.

6. Verification ledger

claim	status
L1, L2, L4 and the theorem	PROVED. L1 and L2 also machine-verified ($x < 200,000$ and $60,000$ respectively); L4 is the induction from L1.
L3 constant $c = 0.5453$	PROVED as an upper bound (one residue class per branch). The measured average 0.5452 is an observation, not part of the proof.
The exact counts in Section 4	MACHINE-VERIFIED and complete by L4. Cross-checked against forward brute-force simulation of every $b \leq 60,000$ — the backward enumeration and the forward scan agree set-for-set, not just in count.
“About 1.7 seeds per octave” and the geometric layer decay	OBSERVED (exactly, at the listed x), not proved. The proved bound is polynomial of degree $L+1$ in $\log x$.
Anything at unbounded depth	OPEN, and equivalent to a single-orbit equidistribution statement of Collatz strength. Nothing here is evidence for it.
The plan’s monotonicity premise	FALSE, corrected here and in NEXT_STEPS.md. Expansion is the property the argument uses.

Code: `density.py` (lemmas, exact preimages, complete depth-graded enumeration, tests including the forward/backward cross-check), `counts.py` (the table). Both runnable standalone.